EXPERIMENT – 4

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Aim: To create Activity Diagram for <case study name>

Prerequisite:

1. Basic understanding of functionality for the system to be designed

Outcome: After successful completion of this experiment students will be able to

1. Understand the various activities in the system to be developed.
2. Identify the actors who would be performing these activities.

Theory:

1) What is Scenario based modeling?

2) How to find out the set of activities in a activity diagram

3) Guidelines for finding out the usage of various symbols in an activity diagram.

Activity diagram is another important diagram in UML to describe dynamic aspects of the system.

Activity diagram is basically a flow chart to represent the flow form one activity to another activity. The activity can be described as an operation of the system.

So the control flow is drawn from one operation to another. This flow can be sequential, branched or concurrent. Activity diagrams deals with all type of flow control by using different elements like fork, join etc.

Purpose:

The basic purposes of activity diagrams are similar to other four diagrams. It captures the dynamic behaviour of the system. Other four diagrams are used to show the message flow from one object to another but activity diagram is used to show message flow from one activity to another.

Activity is a particular operation of the system. Activity diagrams are not only used for visualizing dynamic nature of a system but they are also used to construct the executable system by using forward and reverse engineering techniques. The only missing thing in activity diagram is the message part.

It does not show any message flow from one activity to another. Activity diagram is some time

considered as the flow chart. Although the diagrams looks like a flow chart but it is not. It shows different flow like parallel, branched, concurrent and single.

So the purposes can be described as:

- Draw the activity flow of a system.
- Describe the sequence from one activity to another.
- Describe the parallel, branched and concurrent flow of the system.

How to draw Activity Diagram?

Activity diagrams are mainly used as a flow chart consists of activities performed by the system. But activity diagram are not exactly a flow chart as they have some additional capabilities. These additional capabilities include branching, parallel flow, swimlane etc.

Before drawing an activity diagram we must have a clear understanding about the elements used in activity diagram. The main element of an activity diagram is the activity itself. An activity is a function performed by the system. After identifying the activities we need to understand how they are associated with constraints and conditions.

So before drawing an activity diagram we should identify the following elements:

- Activities
- Association
- Conditions
- Constraints

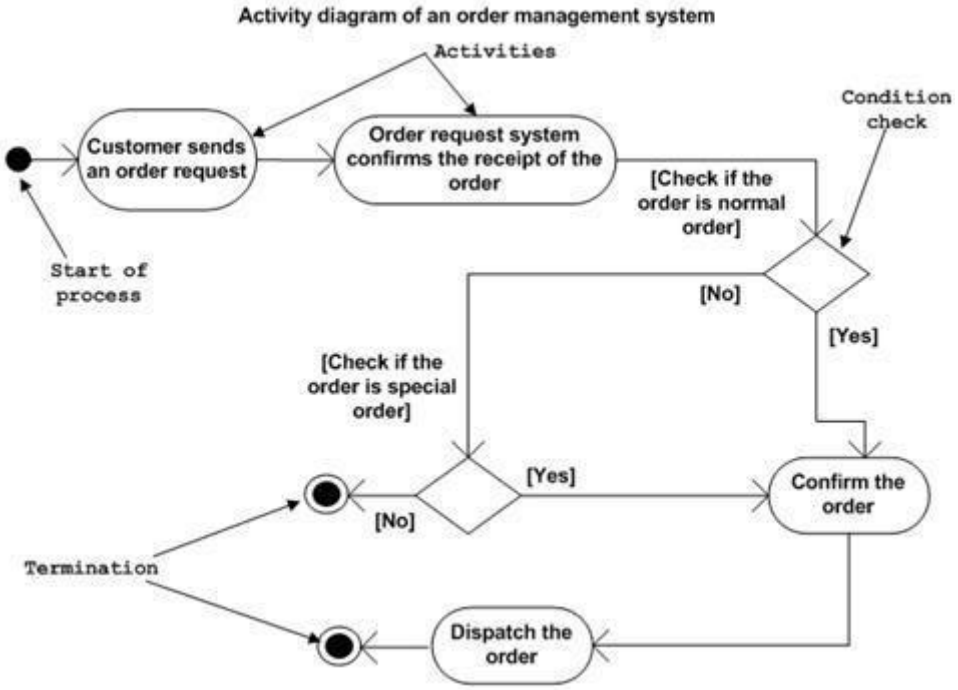
Once the above mentioned parameters are identified we need to make a mental layout of the entire flow. This mental layout is then transformed into an activity diagram.

The following is an example of an activity diagram for order management system. In the diagram four activities are identified which are associated with conditions. One important point should be clearly understood that an activity diagram cannot be exactly matched with the code. The activity diagram is made to understand the flow of activities and mainly used by the business users.

The following diagram is drawn with the four main activities:

- Send order by the customer
- Receipt of the order
- Confirm order
- Dispatch order

After receiving the order request condition checks are performed to check if it is normal or special order. After the type of order is identified dispatch activity is performed and that is marked as the termination of the process.



Where to use Activity Diagrams?

The basic usage of activity diagram is similar to other four UML diagrams. The specific usage is to model the control flow from one activity to another. This control flow does not include messages.

The activity diagram is suitable for modeling the activity flow of the system. An application can have multiple systems. Activity diagram also captures these systems and describes flow from one system to another. This specific usage is not available in other diagrams. These systems can be database, external queues or any other system.

Now we will look into the practical applications of the activity diagram. From the above discussion it is clear that an activity diagram is drawn from a very high level. So it gives high level view of a system. This high level view is mainly for business users or any other person who is not a technical person.

This diagram is used to model the activities which are nothing but business requirements. So the diagram has more impact on business understanding rather implementation details.

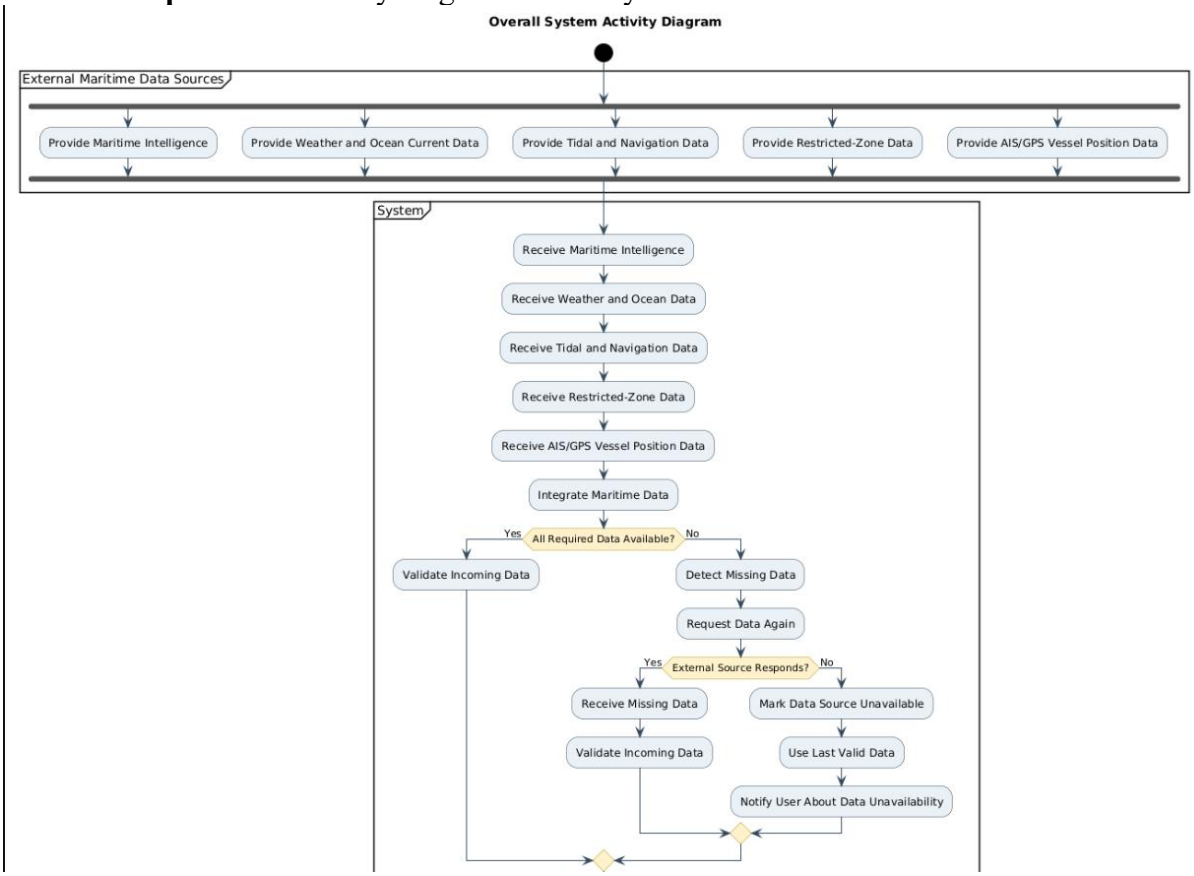
Following are the main usages of activity diagram:

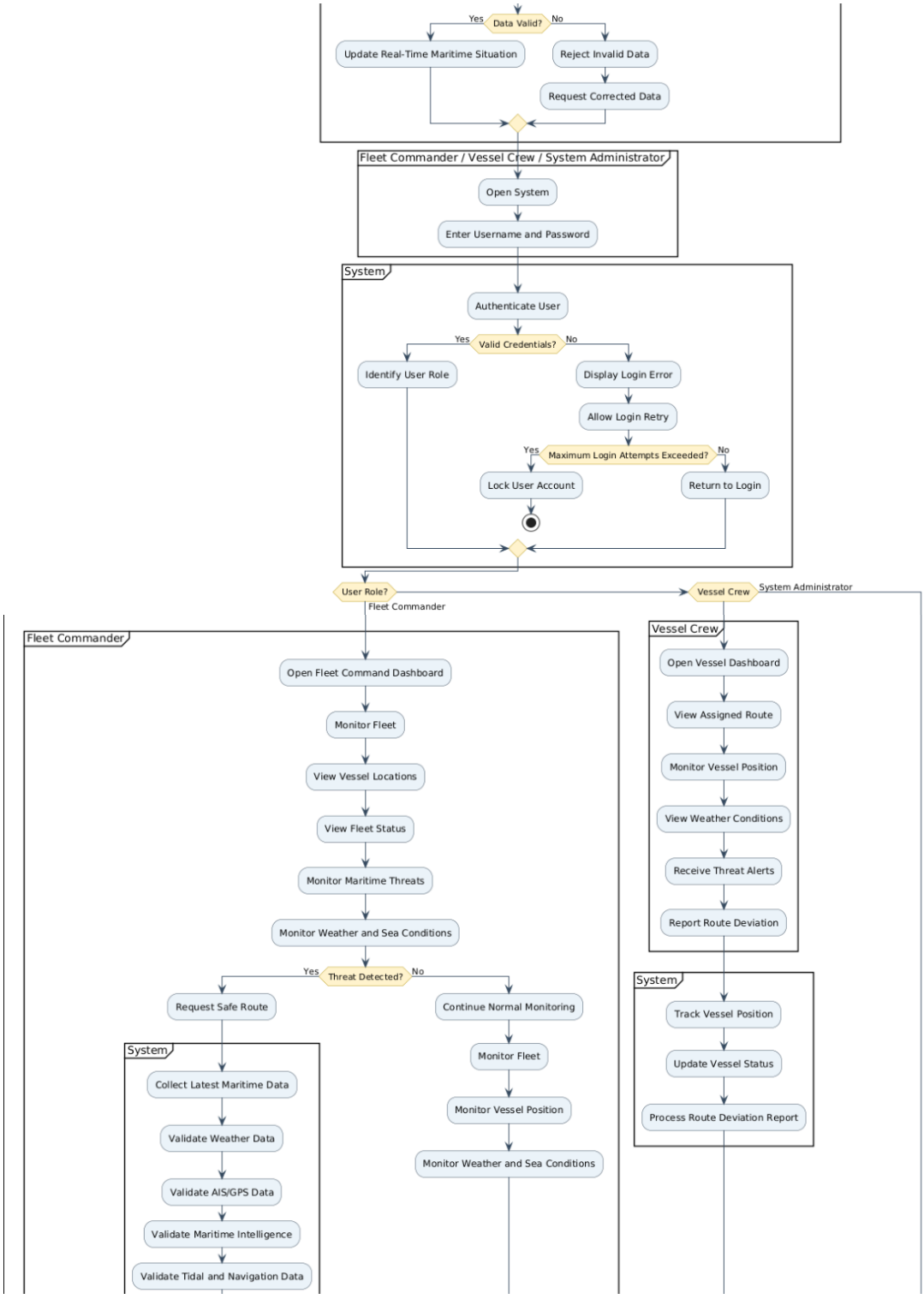
- Modeling work flow by using activities.
- Modeling business requirements.
- High level understanding of the system's functionalities.
- Investigate business requirements at a later stage.

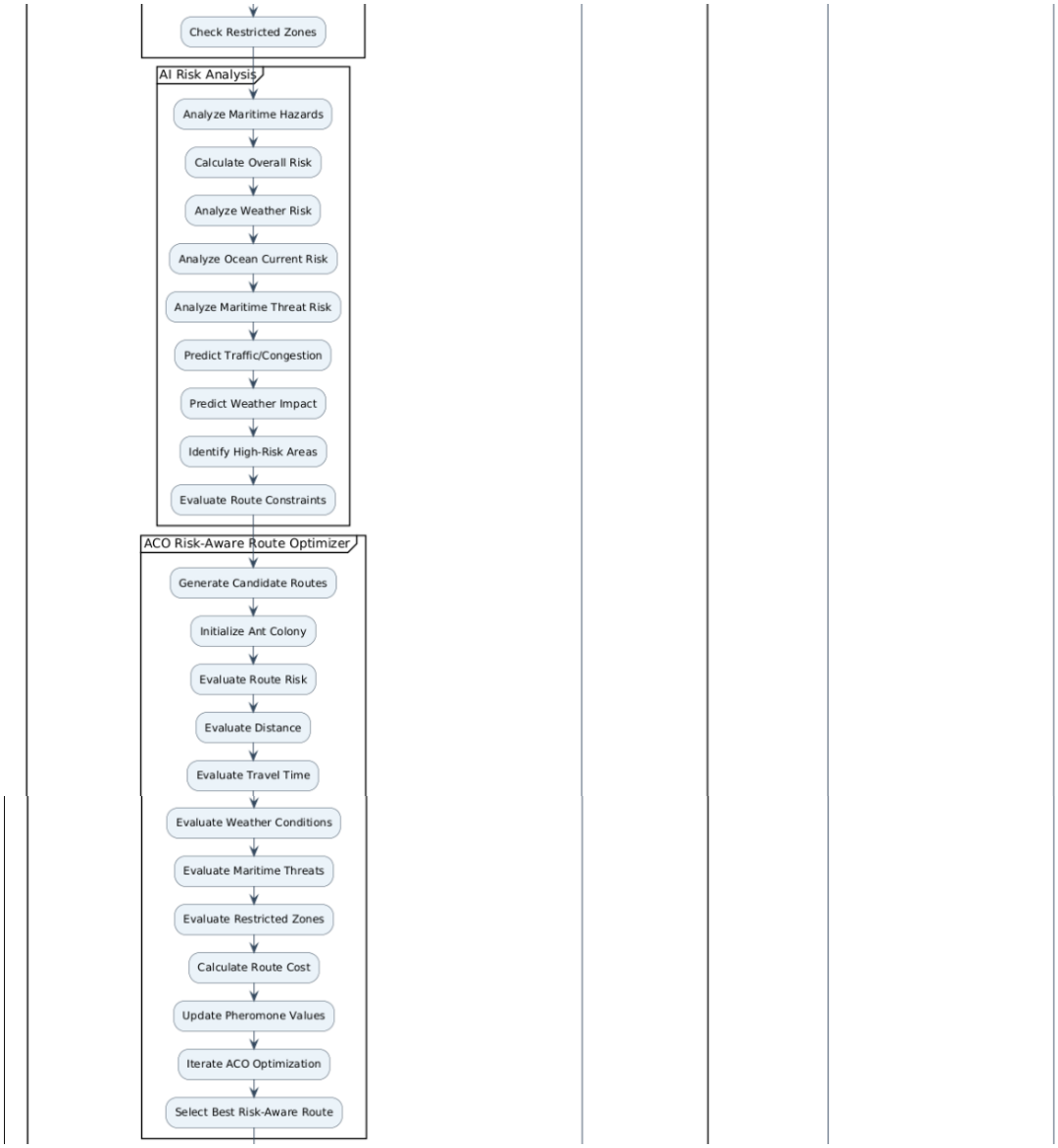
Procedure:

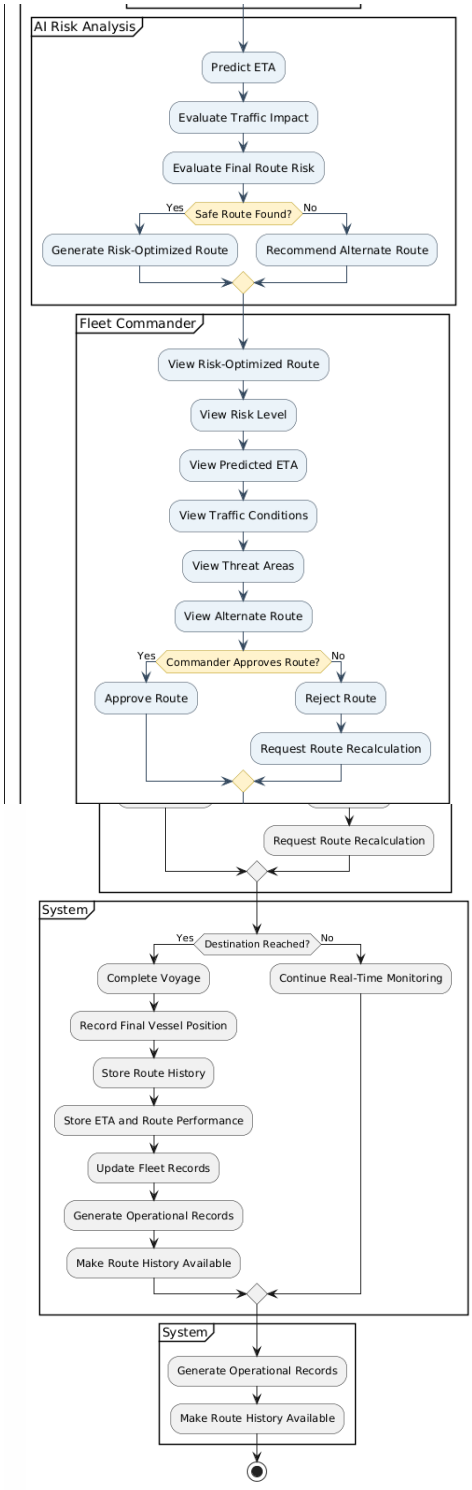
1) Analyze the system to find out the various activities.
2) List down the set of activities and the various Actors in the system who would be performing those set of activities.
3) Draw a Activity diagram showing all the activities.

Output: A Activity diagram for the System.




Yes
Data Valid?
No
Update Real-Time Maritime Situation
Reject Invalid Data
Request Corrected Data
Fleet Commander / Vessel Crew / System Administrator
Open System
Enter Username and Password
System
Authenticate User
Yes
Valid Credentials?
No
Identify User Role
Display Login Error
Allow Login Retry
Yes
Maximum Login Attempts Exceeded?
No
Lock User Account
Return to Login
User Role?
Fleet Commander
Vessel Crew
System Administrator
Fleet Commander
Open Fleet Command Dashboard
Monitor Fleet
View Vessel Locations
View Fleet Status
Monitor Maritime Threats
Monitor Weather and Sea Conditions
Yes
Threat Detected?
No
Request Safe Route
Continue Normal Monitoring
System
Collect Latest Maritime Data
Validate Weather Data
Validate AIS/GPS Data
Validate Maritime Intelligence
Validate Tidal and Navigation Data
Monitor Fleet
Monitor Vessel Position
Monitor Weather and Sea Conditions
Vessel Crew
Open Vessel Dashboard
View Assigned Route
Monitor Vessel Position
View Weather Conditions
Receive Threat Alerts
Report Route Deviation
System
Track Vessel Position
Update Vessel Status
Process Route Deviation Report


Check Restricted Zones
AI Risk Analysis
Analyze Maritime Hazards
Calculate Overall Risk
Analyze Weather Risk
Analyze Ocean Current Risk
Analyze Maritime Threat Risk
Predict Traffic/Congestion
Predict Weather Impact
Identify High-Risk Areas
Evaluate Route Constraints
ACO Risk-Aware Route Optimizer
Generate Candidate Routes
Initialize Ant Colony
Evaluate Route Risk
Evaluate Distance
Evaluate Travel Time
Evaluate Weather Conditions
Evaluate Maritime Threats
Evaluate Restricted Zones
Calculate Route Cost
Update Pheromone Values
Iterate ACO Optimization
Select Best Risk-Aware Route



Explaination:

The activity diagram represents the complete workflow of the **Maritime Fleet Defense & Risk-Aware Route Optimization System**. It starts with receiving maritime intelligence, weather and ocean data, tidal and navigation data, restricted-zone information, and AIS/GPS vessel data, which are integrated and validated by the system. The user then logs in, gets authenticated, and their role is identified as Fleet Commander, Vessel Crew, or System Administrator. The Fleet Commander can monitor vessels, weather, sea conditions, and maritime threats; if a threat is detected, the system performs AI-based risk analysis and uses

Ant Colony Optimization (ACO) to generate and evaluate candidate routes based on risk, distance, travel time, weather, threats, and restricted zones. The safest route is then presented to the Fleet Commander for approval; if rejected, the route is recalculated. Once approved, the system continuously monitors the voyage and updates vessel and route information in real time until the destination is reached, after which the voyage and route records are stored and the process terminates.

Conclusion

The activity diagram successfully represents the overall workflow of the Maritime Fleet Defense & Risk-Aware Route Optimization System, including authentication, monitoring, threat detection, AI risk analysis, ACO route optimization, route approval, real-time updates, and exception handling. It provides a clear view of how the system responds to changing maritime conditions and ensures safer route planning.